PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Methodological quality of multivariate prognostic models for intracranial hemorrhages in intensive care units: a systematic review
AUTHORS	Simon-Pimmel, Jeanne; Foucher, Yohann; Léger, Maxime; FEUILLET, Fanny; Bodet-Contentin, Laetitia; Cinotti, Raphaël; Frasca, Denis; Dantan, Etienne

VERSION 1 – REVIEW

REVIEWER	T Gregório Centro Hospitalar de Vila Nova de Gaia EPE Espinho Unidade III
REVIEW RETURNED	21-Dec-2020

GENERAL COMMENTS	Thank you for the opportunity to review this manuscript. It is well written and its contents are highly relevant. The references are up to date. The methodology behind the manuscript is solid and adequate. The decision to include both intracerebral hemorrhage and subarachnoid hemorrhage is arguable, as the pathophysiology behind both diseases is distinct. However, I recognize that the subject of the manuscript is not pathophysiology but methodological issues for prediction model studies. I have only minor reviews to suggest: - Page 8 (last paragraph of "Study selection")- the authors state that only one reviewer screened the title/abstract. It is generally recommended that this is done by two reviewers to avoid missing eligible studies. This should be acknowledged as a limitation. - Results section: the authors should objectively present the results and refrain from making considerations. Although important and appropriate, these considerations should be moved to the discussion section. Examples of such considerations include: . "The authors should clearly state this information to inform the user of when to calculate the prediction" (page 10) . "Authors who studied long-term outcomes always chose to use logistic regression by excluding patients lost to follow-up when times-to-event methods would have been more appropriate in the presence of such censoring" (page 11) . "This precision allows the reader to assess the risk of overfitting" (page 11) . "Calibration curves, rarely reported, help future external validation assess the eventual need for recalibration or updating to adapt it to the population of interest" (page 11) . "These studies would benefit from external validations with recent and large cohorts" (page 12) . "We this think that only the SAHIT score deserve final attention" (page 14) - Discussion section: on page 16 the authors write "we think that this study deserves attention because it surpasses the functional outcomes uniquely by integrating the quality of life as an objective prediction". This is a very important point. Prediction models mostly

	focus on either death or functional outcome scales and therefore fail to incorporate patient perceptions about the outcome of the disease. I recommend that authors further discuss and emphasize this. Examples of PROM reviews for stroke and subarachnoid hemorrhage include the studies by Reeves (Stroke 2018;49:1549-56) and Nobels-Janssen (Neurology 2019;92(23):1096-1112)

REVIEWER	Aristeidis H. Katsanos McMaster University
REVIEW RETURNED	08-Jan-2021

GENERAL COMMENTS	This is a systematic review with the aim to assess methodologically available prognostic tools for functional outcomes of patients with spontaneous intracranial hemorrhage. Presented information lack novelty and the main outcomes of the current work are already known to most clinicians and researchers handling patients with ICH. 1. Why only "severe" ICH? 2. "We conducted a systematic review of international peer reviewed journals". Review is performed through database searching. 3. Were only primary or also secondary ICH cases included in the studies presented in the review? 4. "Rather than developing new scores, future authors should focus on externally-validating and updating existing scores". Many investigators will argue on this statement, which is a personal opinion of the authors. 5. Validation in specific patient populations of the scores is needed (e.g. anticoagulant related ICH, J Neurol Sci. 2020;409:116628). 6. Table 1 provides no meaningful information, while Table 2 is too large and complex to follow.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1

Dr. T Gregório, Centro Hospitalar de Vila Nova de Gaia EPE Espinho Unidade III

Thank you for the opportunity to review this manuscript. It is well written and its contents are highly relevant. The references are up to date. The methodology behind the manuscript is solid and adequate. The decision to include both intracerebral hemorrhage and subarachnoid hemorrhage is arguable, as the pathophysiology behind both diseases is distinct. However, I recognize that the subject of the manuscript is not pathophysiology but methodological issues for prediction model studies.

I have only minor reviews to suggest:

1. **Page 8 (last paragraph of "Study selection")- the authors state that only one reviewer screened the title/abstract. It is generally recommended that this is done by two reviewers to avoid missing eligible studies. This should be acknowledged as a limitation.**

We agree with this limitation and added this in the discussion section.

> *"This review has several limitations. […] Secondly, only one assessor conducted the study screening on title/abstract. This may have resulted in some missing eligible studies. […]"*

2. **Results section: the authors should objectively present the results and refrain from making considerations. Although important and appropriate, these considerations should be moved to the discussion section. Examples of such considerations include:**

- **"The authors should clearly state this information to inform the user of when to calculate the prediction" (page 10)**
- **"Authors who studied long-term outcomes always chose to use logistic regression by excluding patients lost to follow-up when times-to-event methods would have been more appropriate in the presence of such censoring" (page 11)**
- **"This precision allows the reader to assess the risk of overfitting" (page 11)**
- **"Calibration curves, rarely reported, help future external validation assess the eventual need for recalibration or updating to adapt it to the population of interest" (page 11)**
- **"These studies would benefit from external validations with recent and large cohorts" (page 12)**
- **"We this think that only the SAHIT score deserve final attention" (page 14)**

Following your recommendations, we changed several sentences to report the results objectively. In the discussion section, we modified the paragraph regarding the reviewed studies' limitations in which we brought these methodological considerations.

3. **Discussion section: on page 16 the authors write "we think that this study deserves attention because it surpasses the functional outcomes uniquely by integrating the quality of life as an objective prediction". This is a very important point. Prediction models mostly focus on either death or functional outcome scales and therefore fail to incorporate patient perceptions about the outcome of the disease. I recommend that authors further discuss and emphasize this. Examples of PROM reviews for stroke and subarachnoid hemorrhage include the studies by Reeves (Stroke 2018;49:1549-56) and Nobels-Janssen (Neurology 2019;92(23):1096-1112)**

We thank the reviewer for acknowledging this point, and we agree. As suggested, we clearly discussed this important point, notably expanding the paragraph on the importance of patient's perceptions of its quality of life (QOL) where we cited the PROM reviews (Reeves et al. 2018; Nobels-Janssen et al. 2019).

> *"[…] Moreover, patient perception could weigh the different levels of functional disabilities [50, 51. Indeed, survivors have a wide range of life-long consequences such as neuropsychological difficulties, memory problems, fatigue and physical complaints, i.e. dimensions not explored with functional outcome scales [51,52]. As these symptoms are not always apparent, only validated patient (or caregiver) reported questionnaires can reflect the subjective perception of their quality of life [51,53]. In our systematic review, the only article*

mentioning quality of life concerns the FRESH score [11]. Even though some methodological choices are questionable in this study, we think that it deserves attention because it surpassed the functional outcomes by integrating the quality of life as an objective."

"[…] The use of 'patient-centered' outcomes that have yet to be defined could also enhance the tools in the delicate, medical, and ethical setting of critical care. Beyond all methodological issues, patient-centered clinical finality should guide prognostic tools to be convincing."

Reviewer: 2

Dr. Aristeidis H. Katsanos, McMaster University

This is a systematic review with the aim to assess methodologically available prognostic tools for functional outcomes of patients with spontaneous intracranial hemorrhage. Presented information lack novelty and the main outcomes of the current work are already known to most clinicians and researchers handling patients with ICH.

There is no review of prognostic tools dedicated to severe ICH/SAH managed in ICU to the best of our knowledge. As prognostic scores are only reliable when validated in the population of interest (Moons et al. 2015), it appeared essential for us to conduct this systematic review. In the following answers, you will find detailed explanations that lead us to believe this is an original work.

1. **Why only "severe" ICH?**

Our review aimed to identify prognostic tools used in hemorrhagic strokes admitted to Intensive Care Units (ICU). These patients, who need advanced resuscitation, do not share the same questioning as the population with mild strokes in terms, notably, of pursuing care. Scores developed or validated in unselected ICH patients included mixed populations in terms of severity level. This heterogeneity alters the predictive performance of the scores for specific populations such as severe patients. It is essential to use prognostic tools developed or validated in this specific population. To the best of our knowledge, there is no review of prognostic tools dedicated to ICU or severe patients. Patient admission criteria to ICU are heterogenous, and study designs did not always specify the admission units. We thus chose to include studies focusing on patients with stroke "managed in ICU" or specified explicitly as "severe" or "high grade" injury.

To better justify our choice, we added two sentences in the <u>introduction section</u>.

"[…] This serious statement highlights the initial issues specific to severe strokes and the challenge physicians and surrogates face in deciding to continue invasive care [6,7]. […]"

"Previous reviews of prognostic tools [10-14], popular in the field of neurocritical care, have not focused on injuries managed in ICUs, for whom the issue of advanced care pursuits is a concern. Indeed, scores are reliable when validated in the population of interest. […]"

We completed inclusion criteria in the method section.

"We included all-language studies focusing on adults with severe spontaneous intracranial hemorrhage (ICH or SAH) managed in ICU, or specified explicitly as "severe" or "high grade" injury. We did not include criteria on the location, the cause of the hemorrhage or the type of cases (primary or secondary). […]"

We also added details in the discussion section regarding the limitation related to the definition of these inclusion criteria.

"This review has several limitations. First, we aimed to include tools dedicated to ICH or SAH managed in the ICU. Because of the lack of severity classification for these pathologies, and heterogeneity of patients admitted to ICUs, we defined our proper severity criteria, which is debatable. […]"

2. "We conducted a systematic review of international peer reviewed journals". Review is performed through database searching.

We changed the sentence in the abstract.

"Following PRISMA recommendations, we conducted a systematic review querying Medline, Embase, Web of Science, and the Cochrane database in January 2020."

3. Were only primary or also secondary ICH cases included in the studies presented in the review?

We did not consider any criteria on the type of ICH cases (primary or secondary) for the reviews' study selection. The included studies' inclusion criteria were too heterogeneous to precisely describe the type of case. Most studies did not specify the type of ICH included. Some studies included only primary cases, excluding hematological disorders and malformations. Some excluded partly secondary ICH, as for instance by excluding malformation cases but without excluding patients with anticoagulant or hematological disorders. None solely focused on secondary ICH. The Supplementary Tables present the information regarding inclusion and non-inclusion criteria of each study (Table S3: columns "inclusion criteria" and "non-inclusion criteria", Table S4: "type of injuries").

To describe this point clearly, we completed the inclusion criteria in the methods section and added three sentences in the results section.

> *"[…] We did not include criteria on the location, the cause of the hemorrhage or the type of cases (primary or secondary hemorrhage). […]"*
>
> *"[…] Inclusion criteria were heterogeneous in terms of location or etiology of the hemorrhage. For ICH, most studies included only spontaneous ICH, some excluding malformations and/or coagulation disorders. For SAH, most included aneurysmal SAH. Supplementary Tables 3 and 4 present the information regarding inclusion and non-inclusion criteria of each study. […]"*

4. **"Rather than developing new scores, future authors should focus on externally-validating and updating existing scores". Many investigators will argue on this statement, which is a personal opinion of the authors.**

The prognosis community agrees that there are many prediction models, almost one per day (Moons et al. 2017), most of them only concerning development models (Siontis et al. 2015). However, prediction models perform poorly on external validation studies when authors use suboptimal statistical approaches (Steyerberg 2018; Steyerberg et al. 2018), which is very common, even since the publication of the TRIPOD statement (Zamanipoor Najafabadi et al. 2020). Because we highlighted scores with optimal development approaches, our opinion, supported by precited papers, is that improvement of these scores with more well-designed external validations and updates/recalibration will lead to the expansion of rigorous and unbiased prognostic tools (Van Calster et al. 2017). This will also avoid the publication of multiple developed prediction models that are frequently used in practice (Collins et al. 2014).

To better argue our remarks, we added several references (Van Calster et al. 2017; Steyerberg 2018; Zamanipoor Najafabadi et al. 2020) and completed the discussion section.

> *"[…] Finally, this resulted in a very low number of prognostic tools that seemed methodologically correct and presenting a reasonable prognostic performance level. However, weaker validation results do not mean that the model is incorrect. If the scores' development was optimal, relevant predictors could be recalibrated and combined with new data to validate a strong tool [57, 58].*
>
> *[…] Similar to Zamanipoor Najafabadi et al., we noticed a trend towards quality improvement, reinforced with the necessary ongoing validation of existing scores [59]."*

5. **Validation in specific patient populations of the scores is needed (e.g. anticoagulant related ICH, J Neurol Sci. 2020;409:116628).**

We added this recommendation in the discussion section and the following references (Appelboom et al. 2012; Katsanos et al. 2020).

> *"[…] We also did not find tools dedicated to severe specific populations (such as hemorrhages secondary to malformation or patients with coagulation disorders). External validations would be interesting for these populations (e.g., patients under anticoagulant [63] or arteriovenous malformation [64]). Another option would be to extend existing scores with these risk predictors, such as the Max ICH-score, which includes the variable "presence of oral anticoagulant" [25]. […]"*

6. **Table 1 provides no meaningful information, while Table 2 is too large and complex to follow.**

One aim of our systematic review was to assess the methodological quality of prognostic scores. We intended to suggest a progression to follow in assessing the robustness of a prognostic tool in the most didactic way. In our view, methods used to quantify the performance of the models are an essential part of this assessment. We proposed to use that table to summarize quickly the type of methods used and their popularity. We understand, though, that from a clinical point of view, Table 1 seems unnecessary. We have moved this to the Supplementary Web Material, becoming Table S5.

To clarify former Tables 2 and 3, we merged some columns and separated some others. We also split former Table 3 in two tables (Table 2 for ICH, Table 3 for SAH).

> *"[…] Table 2 and 3 summarize, for ICH and SAH respectively, the collected information regarding the source population, development approach, validation details, and prognostic performances of these eight scores."*

VERSION 2 – REVIEW

REVIEWER	T Gregório Centro Hospitalar de Vila Nova de Gaia EPE Espinho Unidade III
REVIEW RETURNED	02-May-2021

GENERAL COMMENTS	I am satisfied with the author´s responses.

REVIEWER	Aristeidis H. Katsanos McMaster University
REVIEW RETURNED	22-Apr-2021

GENERAL COMMENTS	Authors responded successfully to all comments.